

How the model was developed and tested

01

Planned SST test

Full neuron-specific responses

Files **9 SST-to-E**

Single-neuron trials **26,803**

Eligible group trials **19,217**

MSE **q = 0.00391 pass**

Waveform rho **q = 0.682 no gain**

MSE GAIN | NO SHAPE GAIN

02

Develop rule in SST

Reliability from trial halves

Files **9 SST-to-E**

Eligible group trials **19,217**

Mean reliability **0.781**

MSE **q = 0.00391 pass**

Waveform rho **q = 0.0469 pass**

RULE LOCKED BEFORE NEW DATA

03

Held-out E-to-E cohort

Same rule, no tuning

Files **10 E-to-E**

Single-neuron trials **129,613**

Eligible group trials **74,435**

MSE **q = 0.00195 pass**

Waveform rho **q = 0.00879 pass**

FILE-LEVEL TEST PASSED

Datasets are not pooled for significance; the E-to-E result cannot revise the original SST/PV decision.